DGPO inside πRL

Replacing Flow-SDE GRPO with Direct Group Preference Optimization
for flow-based VLAs ($\pi_{0.5}$)

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The question

- Plan from the postdoc: pick an RL method (Flow-GRPO, DGPO, Prism-GRPO), put it into πRL, and post-train π_0 / $\pi_{0.5}$.
- **Idea 1 (Euclidean):** keep πRL's model, environments, rollouts and reward; **replace only the Flow-GRPO update with DGPO** (arXiv:2510.08425).
- DGPO's claim: no stochastic policy needed, deterministic ODE rollouts, $\sim$20–30$\times$ faster than Flow-GRPO on image generation.
- Stage 2: swap the binary success reward for Prism-GRPO's $R = \text{success} + \lambda q$ (arXiv:2608.17423).

Answer in one line: in every setting we tested, πRL's GRPO improves the policy more and faster than DGPO; Prism adds nothing measurable.

What is what: the components of a πRL experiment

Component	What it is	πRL options	**Used here**
Model (policy)	VLA that outputs actions	π_0, $\pi_{0.5}$	$\pi_{0.5}$
SFT data	demos used before RL	per benchmark (next slide)	RLinf's released SFT checkpoints
Environment	simulator the policy acts in	LIBERO, ManiSkill, MetaWorld, CALVIN	all four
Action	command sent to the robot each step	4–7 dims (next slide)	benchmark default
RL algorithm	update rule from rewards	PPO (main), GRPO	**GRPO, DGPO**
Exploration noise	how the flow sampler is made stochastic	Flow-SDE, Flow-Noise	Flow-SDE; DGPO: ODE or SDE
Reward	score of a trajectory	binary success	binary; **Prism**

Benchmarks: environment, action, data, reward

	LIBERO	ManiSkill	MetaWorld	CALVIN
Robot	Franka Panda	WidowX (Bridge)	Sawyer	Franka Panda
Tasks	4 suites $\times$ 10 (Spatial, Object, Goal, Long)	16 objects $\times$ 17 receptacles $\times$ 16 scenes	MT50: 50 tasks	chains of 5 subtasks; train ABC, eval D
Action	7-d: Δpos 3, Δrot (axis-angle) 3, gripper	7-d: ΔEEF pose 6, gripper	**4-d**: Δxyz 3, gripper	7-d: Δpos 3, Δrot (Euler) 3, gripper
SFT data	few-shot demos; 40 trajectories, one $\pi_{0.5}$ checkpoint for all suites	16,384 MPLib motion-planning episodes	official set, 2,500 trajectories	CALVIN ABC play data ($\sim$24 h)
Reward	success 1 / 0	1.0 success + 0.1 grasp	success 1 / 0	subtask success
Episode	240 steps (Long 480)	80	100	480 / subtask
SFT (ours, noise-free)	0.806 (Spatial), 0.274 (Long)	0.447	0.461	0.136

πRL's own variants and what we ran

πRL paper, Table 1 (PPO), success %

		LIBERO	MS	MW	CALVIN
π_0	SFT	57.6	38.4	50.8	57.5
	Flow-SDE	96.1	78.8	78.1	61.7
	Flow-Noise	97.6	77.8	85.8	59.9
$\pi_{0.5}$	SFT	77.1	40.1	43.8	61.3
	Flow-SDE	97.9	90.9	70.7	87.0
	Flow-Noise	98.3	89.7	66.1	84.5

GRPO is reported only on LIBERO (Table 4, $\pi_{0.5}$ avg. 91.5).

Our runs (all $\pi_{0.5}$, 12 h each)

	Sp.	Long	MW	MS	CV
GRPO + Flow-SDE	✓	✓	✓	✓	✓
DGPO, ODE	✓	✓			
DGPO + SDE	✓		✓	✓	✓
DGPO + SDE + Prism	✓		✓	✓	✓
DGPO, 1-step noise			✓	✓	

Our absolute numbers sit below the paper's: a 12 h budget is 20–200 steps on 4×H100, the paper trains longer on 8.

The training loop (RLinf), and where DGPO plugs in

1. **Rollout:** env workers step the simulator; rollout workers sample action chunks from the flow policy. Groups of 8 envs share one reset state.
2. **Reward & advantage:** binary success per trajectory $\rightarrow$ GRPO-normalised A_i within each group (or Prism's RLOO advantage).
3. **Update (the only part we replace):**
   - GRPO: clipped ratio of the Gaussian transition log-probs at one noised denoise step.
   - DGPO: forward-noise the clean chunk x_0, $x_t = (1-t)x_0 + t\epsilon$; score it with the DSM loss of the policy and of a frozen SFT reference; group-level sigmoid.
4. **Held-out eval:** noise-free (ODE) sampler, 512 episodes per checkpoint (CALVIN: 50–200), same config for every method.

Reproducing a run

Code: ~/pirl-dgpo/RLinf (branch dgpo): `rlinf/algorithms/dgpo.py`, `workers/actor/fsdp_dgpo_policy_worker.py`, configs `*_dgpo_openpi_pi05.yaml`, `*_grpo_openpi_pi05.yaml`.

```
# DGPO on MetaWorld (SDE 0.3 on every denoise step, 2 updates/rollout)
sbatch --export=ALL,CONFIG=metaworld_50_dgpo_openpi_pi05,\
  CKPT=$L/checkpoints/RLinf-Pi05-MetaWorld-SFT,EXTRA_PYTHONPATH=$L/envpkgs/metaworld \
  jobs/01_dgpo_libero_spatial.sbatch

# ...with GRPO's exact exploration noise (one random step, 0.5)
  ... +actor.model.openpi.nft_single_noise_step=True actor.model.openpi.noise_level=0.5

# ...with the Prism reward
  ... algorithm.filter_rewards=False +algorithm.dgpo_reward=prism +algorithm.prism_lambda=0.5

# GRPO control arm
sbatch --export=ALL,CONFIG=metaworld_50_grpo_openpi_pi05,... jobs/02_pirl_grpo_baseline.sbatch

# Held-out eval of any checkpoints (noise-free, 128 envs x 4 epochs)
sbatch --export=ALL,EVAL_CONFIG_DIR=...,EVAL_CONFIG=metaworld_50_dgpo_openpi_pi05_eval,... \
  jobs/03_eval_ckpts.sbatch sft /path/to/global_step_20/actor/model_state_dict/full_weights.pt
```

Configs are generated from πRL's PPO configs by `jobs/make_dgpo_configs.py` / `make_grpo_configs.py`: only the algorithm, the sampler and the training env's reset change.

Cluster notes (Expanse, 4×H100 per node)

- One Python env (`rlinf_venv`); MetaWorld and CALVIN deps are side-installed with `--target` and added through `EXTRA_PYTHONPATH` (MetaWorld needs gymnasium 1.x, which would break LIBERO).
- ManiSkill's Vulkan renderer works only on some nodes (exp-19-15/16/17); others fail with `ErrorDeviceLost` at camera creation.
- CALVIN's EGL renderer SIGABRTs mid-rollout; we use pybullet's CPU renderer (`CALVIN_USE_EGL=0`).
- Every job gets its own Ray port block and temp dir (shared nodes); no `ray stop`.
- LIBERO eval on Long needs the robosuite horizon raised (`+env.eval.init_params.horizon=3000`).
- Budget used: ∼1.5k of 10k H100-hours.

DGPO, and how it maps onto an MDP

DGPO (Eq. 17) for a group $\mathcal{G}$ sharing a condition c:

$$\mathcal{L} = -\log \sigma\Big(-\beta \sum_{x \in \mathcal{G}} A(x) \left[\mathcal{L}^{\theta}_{\text{dsm}}(x) - \mathcal{L}^{\text{ref}}_{\text{dsm}}(x) \right] \Big), \qquad \sum_{\mathcal{G}^+} |A| = \sum_{\mathcal{G}^-} |A| \Rightarrow Z(c) \text{ cancels.}$$

- Images: sample = one image, c = the prompt.
- πRL: every chunk sees a different observation, so no two chunks share c. What a group shares is the **reset state**.
- $\Rightarrow$ **sample** = **trajectory**: $\log \frac{\pi_\theta(\tau)}{\pi_{\text{ref}}(\tau)} = \sum_k \log \frac{\pi_\theta(a_k|o_k)}{\pi_{\text{ref}}(a_k|o_k)}$ (dynamics cancel); each chunk term $\rightarrow$ its DSM difference, averaged over the trajectory's valid chunks.
- A principled trajectory-level extension, not the paper's theorem (the closed form needs deterministic dynamics, which MuJoCo has).

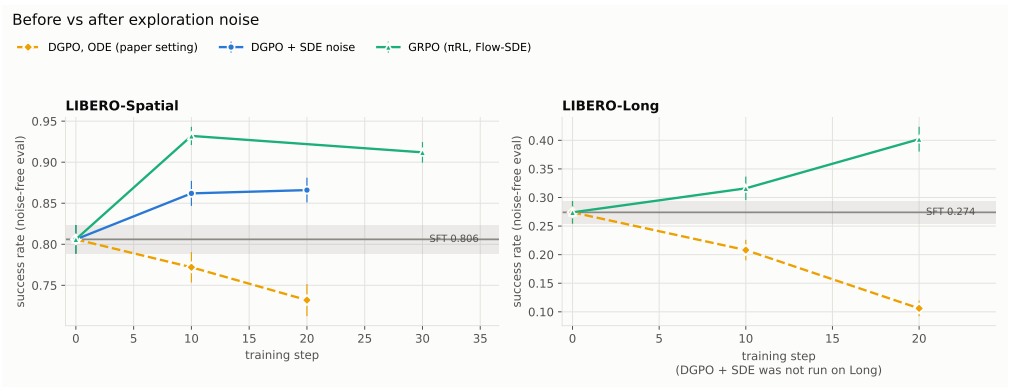
Implementation and verification

- The sigmoid spans all chunks of all trajectories in a group ($\sim 8 \times 48$ forward passes), so the loss cannot be micro-batched, but its gradient can:

$$\partial \mathcal{L}_g / \partial \mathcal{L}_{ik}^{\theta} \propto \sigma(-z_g)\, A_i \;\Rightarrow\; \text{no-grad pass for } z_g, \text{ then backprop } \sum \sigma(-z_g) A_i \mathcal{L}_{ik}^{\theta}.$$

- Unit tests (15): surrogate gradient = autograd of the exact objective; Z cancels; masking; length balance.
- **Matches the official code:** our gradient equals the gradient of `compute_group_dgpo_loss_allreduce` (read from Luo-Yihong/DGPO) for $\beta \in \{1, 10^2, 10^4\}$.
- Two bugs found and fixed on the way: summed (not length-normalised) chunk terms made long failures dominate and collapsed training to 0%; metric averaging over idle ranks.

Before noise: the paper's setting (ODE rollouts) makes the policy worse

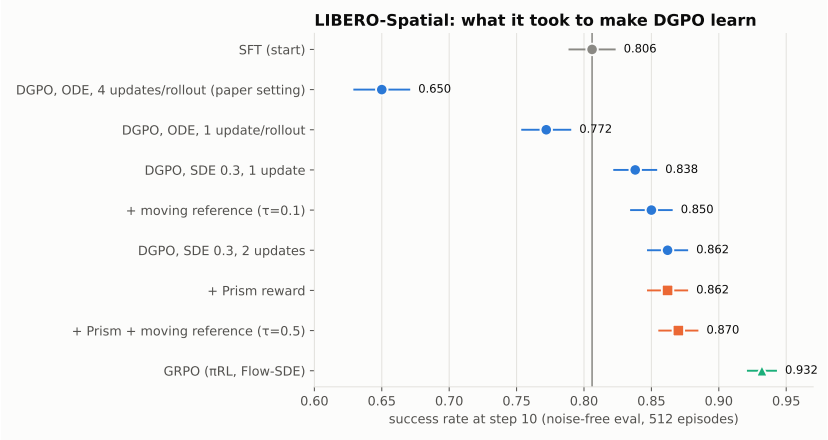


Noise-free held-out success, 512 episodes, ± 1 s.e. Step 0 = SFT. ODE DGPO falls below SFT on both LIBERO suites and keeps falling.

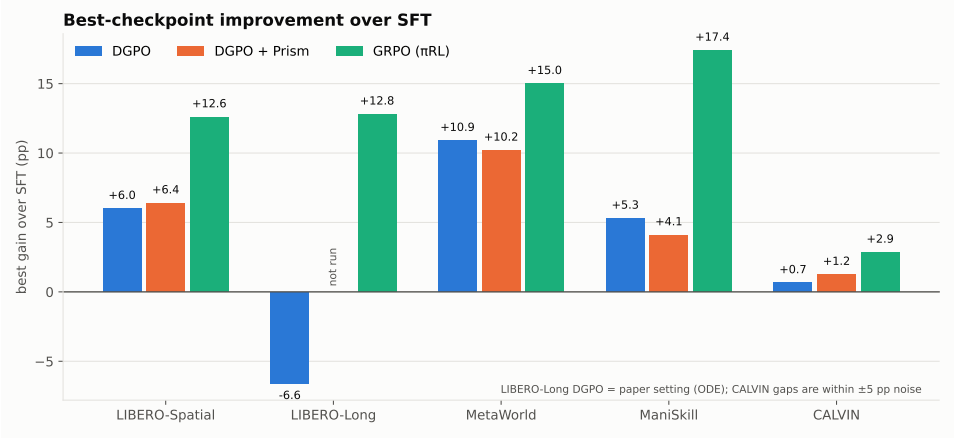
Why the noise is needed

- With ODE rollouts the 8 trajectories of a group differ only in their initial noise. The SFT policy is already strong ($\sim$0.88 on Spatial), so they behave almost alike: only $\sim$40% of groups have mixed outcomes.
- Pushing the likelihood of a failed trajectory down also pushes down the many chunks it shares with successful ones.
- A step-matched SFT control ($\text{lr} = 0$) showed the early "drops" were partly reset-state difficulty, so every run is compared on held-out evals, never on training success.
- Adding SDE noise (DGPO's own "w/ SDE" variant) raises mixed groups to $\sim$90%, and DGPO starts to learn.

What it took to make DGPO learn (LIBERO-Spatial, step 10)

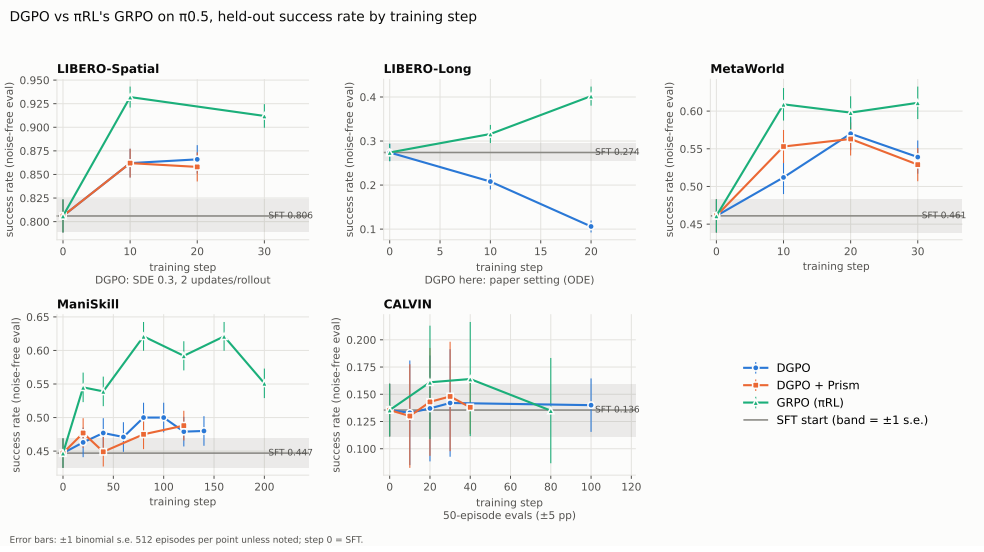


After noise: best improvement over SFT, all benchmarks

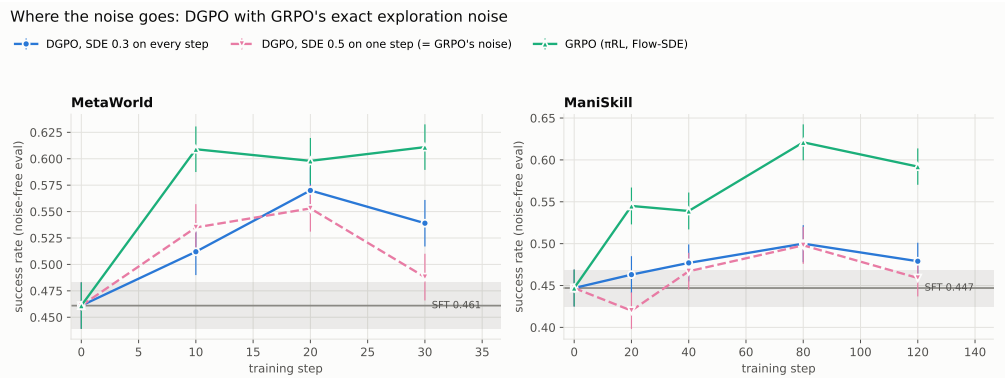


DGPO = SDE 0.3 on every denoise step, 2 updates per rollout, $\beta = 10^4$ (LIBERO-Long: ODE only).

Learning curves

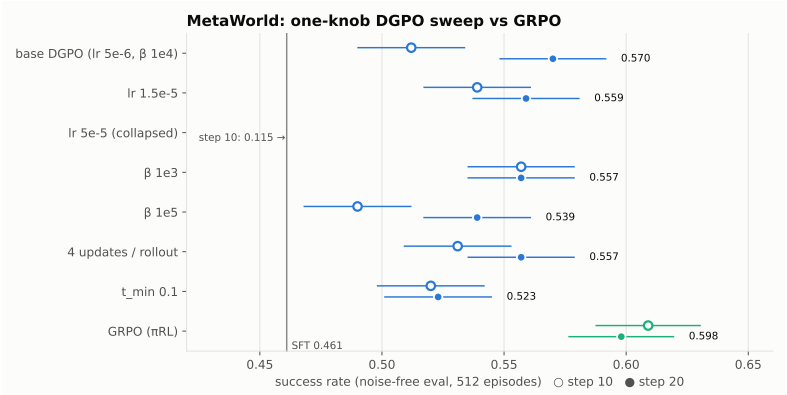


Is it the noise? DGPO with GRPO's exact exploration noise



New option `nft_single_noise_step`: SDE noise 0.5 on one random denoise step, exactly as GRPO's rollouts. Step-1 training success matched GRPO's (MetaWorld 0.438 vs 0.422). The gap remains and the run is less stable, so **the gap is the update rule, not the noise**.

Is it tuning? One-knob sweep on MetaWorld



Nothing beats the base setting at step 20; every setting that learns plateaus at 0.52–0.57.

Stage 2: the Prism reward

- $R = \text{success} + \lambda q$, $\lambda = 0.5$, $q = \exp(-J/\text{median}\, J)$ with J the jerk of the executed continuous action dims (gripper excluded); RLOO advantage (zero-sum, so Z still cancels); group filtering off.
- Result: no measurable gain anywhere (MetaWorld 0.563 vs 0.570, ManiSkill 0.488 vs 0.500, LIBERO-Spatial 0.862 vs 0.866, CALVIN within noise).
- Why: with SDE rollouts $\sim$90% of groups are already mixed, so Prism's tie-breaking of same-outcome groups has little to act on (3–9% of groups).

Why DGPO does not transfer to πRL

1. **The speed-up has nothing to remove here.** The 20–30$\times$ comes from (i) long SDE sampling and (ii) training along the whole denoising chain in Flow-GRPO. πRL rollouts are simulator-bound (3–4 denoise steps), and its Flow-SDE already trains one random step. Measured: GRPO steps were 15–35% *faster* than DGPO's.
2. **ODE rollouts carry no signal** once SFT is strong (previous section).
3. **Early plateau everywhere**, even with the gate kept open ($\beta = 10^3$): a DSM-difference preference over a 50–100-chunk trajectory with one binary reward is a weak, noisy signal compared with GRPO's per-transition likelihood ratio.
4. **DGPO regresses onto the sampled x_0.** With large one-step noise the successful samples carry that noise, and the noise-free policy degrades (ManiSkill step 20: 0.420 $<$ SFT); GRPO optimises the noised transition's likelihood instead.

Caveats

- One seed per configuration (the gaps are large: 4–12 pp at 512 episodes, $\pm$2.2 pp s.e.).
- The trajectory-level extension is ours; DGPO is validated on single-sample generation.
- DGPO sweep was one knob at a time on MetaWorld; no LoRA, no combined tuning.
- CALVIN was scored with `success_once`, not the paper's average chain length.
- Absolute numbers are below πRL's (shorter training, 4 GPUs); we did not run PPO or π_0.
- Training success (with exploration noise) is not a proxy for noise-free success:
  DGPO-1step led GRPO in training success at ManiSkill steps 10–12 and trailed it in eval.

Conclusions

- Replacing πRL's Flow-SDE GRPO with DGPO, as planned, **does not beat GRPO** on any of LIBERO, MetaWorld, ManiSkill or CALVIN.
- The paper's ODE setting degrades the policy; with SDE noise DGPO improves on SFT (+6 / +11 / +5 pp on LIBERO-Spatial / MetaWorld / ManiSkill) but GRPO does more (+13 / +15 / +17 pp).
- Not an implementation issue (matches the official gradient), not tuning (sweep), not the noise (identical rollouts): **the update rule**.
- Prism's reward adds nothing on top of DGPO in these benchmarks.

Possible next steps: DGPO + SDE on LIBERO-Long; π_0; CALVIN with the paper's chain-length metric; Prism on GRPO for completeness (it is the Prism-GRPO paper's own setting).

Backup: all held-out numbers (noise-free success)

Benchmark (SFT)	DGPO + SDE	DGPO + Prism	DGPO 1-step noise	GRPO
LIBERO-Spatial (0.806)	.862 / .866 (s10/20)	.862 / .858	–	.932 / .912 (s10/30)
LIBERO-Long (0.274)	ODE: .208 / .106 (s10/20)	–	–	.316 / .402
MetaWorld (0.461)	.512 / .570 / .539 (s10/20/30)	.553 / .563 / .529	.535 / .553 / .488	.609 / .598 / .611
ManiSkill (0.447)	.463/.477/.500/.479 (s20/40/80/120)	.477/.449/.475/.488	.420/.467/.498/.459	.545/.539/.621/.592
CALVIN (0.136)	.133/.137/.142 (s10/20/30), .140 (s100)	.130/.143/.148, .138 (s40)	–	.161/.164/.135 (s20/40/80)

512 episodes ($\pm$2.2 pp) except CALVIN: 50 episodes ($\pm$5 pp); CALVIN s100 / s40 and SFT: 200 episodes. Full log: `~/pirl-dgpo/README.md`.